Listing 3.5 A simple implementation of a standard linear Kalman filter in one dimension. The example assumes a steady state temperature, and only estimates a single state.

```
import numpy as np
import matplotlib.pyplot as plt

true_temp = 22.0 #1
timesteps = 50 #2

np.random.seed(42)
sensor_noise = [np.random.normal(0, 0.5, timesteps) for _ in range
(5)]
sensors = [true_temp + n for n in sensor_noise] #2
measurements = np.mean(sensors, axis=0) #3

x_est = 20.0 #4
P = 1.0
R = 0.25
Q = 0.01

estimates = []

for z in measurements:
  x_pred = x_est #5
  P_pred = P + Q

  K = P_pred / (P_pred + R) #6
  x_est = x_pred + K * (z - x_pred)
  P = (1 - K) * P_pred

  estimates.append(x_est)

plt.plot(measurements, label="Average Sensor Reading", linestyle="dot
ted")
plt.plot([true_temp] * timesteps, label="True Temperature", linestyle
="--")
plt.plot(estimates, label="Kalman Filter Estimate")
plt.xlabel("Time step")
plt.ylabel("Temperature (°C)")
plt.legend()
plt.show()
```

#1 Assume a constant real temperature of 22.
#2 Generate 50 readings for each of the 5 sensors, assigning random noise to each reading.
#3 Calculate the average from each of the 5 sensor readings.